

Progettazione di Applicazioni Mobili

Mobile app design

Envision – Part 1/2

Stefano Burigat

Dipartimento di Scienze Matematiche, Informatiche e
Fisiche - Università di Udine

stefano.burigat@uniud.it



- Different techniques/tools can be used to support the envision stage:
 - System concept statement
 - Learning plan
 - Guerrilla user research
 - Competitive/comparative assessment
 - Storyboarding
 - User personas
 - Requirements description
- Not all of these techniques/tools are necessary but the more you use the less uncertainty there is in later steps of the process

System concept statement

- A system concept statement is a concise descriptive summary of the envisioned application
- A clear system concept statement is important because it
 - helps to explain the system to outsiders
 - acts as a baseline for reality checks and product scope and as something to point to in the middle of later design

- A system concept statement is typically 100 to 150 words in length (but can be longer and more detailed)
- An effective system concept statement answers at least the following questions
 - What is the system name?
 - Who are the system users?
 - What problem(s) will the system solve?
 - What will the system do?
 - What experience will the system provide to the user?

- A system concept statement is a live description
- You will start with a short statement that states your initial idea of the application
- There will be uncertainty associated to this initial idea
 - You won't likely be able to clearly answer all questions associated to the statement
- You need to get information to reduce such uncertainty and refine your statement

The Ticket Kiosk System will replace the old ticket retail system, the Middleburg University Ticket Transaction Service, by providing 24-hour-a-day distributed kiosk service to the general public. This service includes access to comprehensive event information and the capability to rapidly purchase tickets for local events such as concerts, movies, and the performing arts.

The new system includes a significant expansion of scope to include ticket distribution for the entire MU athletic program. Transportation tickets will also be available, along with directions and parking information for specific venues. Compared to conventional ticket outlets, the Ticket Kiosk System will reduce waiting time and offer far more extensive information about events. A focus on innovative design will enhance the MU public profile while Fostering the spirit of being part of the MU community and offering the customer a Beaming interaction experience. (139 words)

- Supported by the Yonsei Residential College, Guess Who? Is a location-based smartphone app that pairs up YIC students based on common interests. First, students will download the app and register as users by entering their Yonsei portal ID and password. Then they will take a survey about their basic information (name, gender, age, RC/non-RC, hobbies). After submitting, students will be paired up based on common interests, and they will start a game to try and find out their partner's identity.

Concept

The weather app for weather nerds.

Details

The app will include all detailed feeds available from national weather services in an attractive interface with weekly, daily and hourly views.

Customer Needs

The most popular weather apps offer minimal information such as probability of precipitation and temperature ranges per day. This is lacking for many purposes such as planning an event whereby precise hour by hour forecasts are often required. There are also a significant number of users who are nerdy about weather such that they crave details.

"Fishing Wichita" App Definition	
What does the app do? (<i>Summary</i>)	The app will gather weather/water conditions and fishing reports from popular fishing lakes and rivers in a 150 mile radius of Wichita and combine them into one easy to read and scroll display.
What exactly does the app do? (Technical Description)	Using available on-line data or information, it will collect conditions and then consolidate them into an easy to read display. The app will update conditions to user settings, such as every morning or every Saturday morning. Conditions can also be updated upon request. Users can enter set-up information to customize app in terms of update frequency, and site categories (i.e. favorites, complete list, ignore).
Why would someone want the app?	It is time consuming to gather current conditions information about area lakes and rivers, but without information choosing the best place to fish is just a guess. Time is precious when leaving to fish, especially in terms of getting an early start. Having quick and easy access to all the area lake and river conditions would have value for people who like to fish.
Who would want the app?	The market or audience would be people in the Wichita area who fish regularly in area waters.

APP NAME | Catchy, short, and relevant

B.Y.O.Bike

CATEGORY | Business, entertainment, finance, etc.

Business, personal

CONTEXTS, BEHAVIORS, EMOTIONS | Themes underlying the concept

Any user can build his/her own virtual motorcycle from the frame on out.

DESCRIPTION | What does it do?

B.Y.O.Bike allows users to select images of motorcycle components and visually place them together as a whole unit. Components can be easily switched out, paint finishes can be changed, and costs can be calculated.

VISUALIZE THE IDEA | Using words or drawings

For any D.I.Y. motorcycle builder, sourcing the variety of components can be a challenge. B.Y.O.Bike collects parts on-the-fly from original equipment manufacturers as well as after market companies. The app will be well-designed with core functionalities that will be easy to use. Well-photographed images of shiny products will entice the user at every phase of building the bike. With a few gestures on this app, the user can create a customized bike, built with his own two fingers. After the user is finished with the build, a rough cost can easily be tallied.

INTERACTION AND ENGAGEMENT | How does it work? What is the user flow? How is it engaging?

The user first selects a frame style—cruiser, street, chopper, touring, dirt—whatever he/she is interested in. The app provides the next logical component to select until the last part is in place. Paint and metal finishes can be sampled along the way—colors, metallic/non metallic, polished/satin chrome, gloss/matte black, etc. Or, the user can use the app's image recognition capability to capture any available part within sight, and let the app bring up related components. Direct links to manufacturers will allow the user to research each part as it is being virtually built. The user can share his ideas with friends and get feedback, making it a very fun and social event. For additional visual effects, a variety of backgrounds such as roads, diners, and garages could be selected to be placed behind the bike so it lives in an environment. After the user is finished, he selects from a number of builders advertised on the app. He sends his composite to the builder's shop and begins the collaborative refinement process of the build. During the actual building process, components and parts are easily ordered, as they are already stored in a shopping cart.

SwiftShopper is a location-aware mobile app that will help its users shop smarter by providing a faster shopping experience and help users make better-informed shopping decisions. SwiftShopper will include an interactive shopping list for the shopper. The user will easily be able to look up any item located in the store using the search functionality, and the employee will be able to add or update the store's inventory of items. Once the user completes his or her shopping list, it will be possible to view a path that covers all of the items in the list using a map. As a result of SwiftShopper's features, the user will have a vastly improved shopping experience, which will also be beneficial to the store in that sales are easier to make when the user is happy. Finally, the user will be less likely to forget their items and become stressed because they will be able to use the intelligent shopping list feature of our app if they choose to do so.

Learning plan

- A learning plan is a structured approach that can be used for vision-definition and early concept exploration when you start a new project
- It helps you move from uncertain assumptions to validated knowledge by systematically identifying:
 - what you need to learn
 - how you will learn it
 - how those learnings will inform your next steps

- Start with what you know
 - list everything you think you know (your working assumptions) about your users, needs, contexts, or feasibility
 - Who are the intended users, and what is their context of use? What user needs or pain points are we trying to understand or solve? How do users carry out their tasks? What features and functions are most important to them? Do they use other products?

ASSUMPTIONS ABOUT OUR USERS

TYPES OF USERS →	VIEWERS	ADMINS	BUYERS
GOALS/ STATE OF MIND	- Search + find correct answers - Want to be certain - Want to share what they find with others - Want to be good at their jobs/school	- Dealing with piles/lots of docs - Provide good svc. - Solve problems. Repeat processes "Add Value"	- Save & ensure feature parity - Pleasure their internal customers - Create automated processes
PAIN POINTS	- Not sure if they've found the only/all the info - Don't save + duplicate portfolios - Can't customize with their name/logo - Not enough time	- Ability to copy/cut close a doc in a handy fashion - Listen for keywords - Creation of repeatable scripts for common issues	- Not aware of all the service options - Don't necessarily discriminate b/w... not everyone knows what they can do - Rigid systems
OTHER PRODUCTS THEY USE	Excel Word Dropbox Paper + Pen Sticky Notes Binders + Photographs Evernote	Sharepoint MS Access Email Archives Colleagues Internet Evernote	Analyst Reports Procurement System Office Email
COPING STRATEGIES	Email colleagues Create duplicates/clones Keep notes/files Bookmarks	Support Scripts Every time Permission to do things Relationship with other admins	Talk to colleagues at other places Look for good alternatives in other people's needs

- Separate certainties from assumptions
 - for each assumption, indicate how confident you are in that assumption
 - focus on the assumptions that carry the highest risk
- Design and conduct experiments or research
 - for each of your questions, define how you might go about getting more data or info in this area (e.g., interviews, market research, diary studies, concept tests, competitor benchmarking)
 - think about what form your evidence will take
- Document insights and evolve the vision statement after each learning cycle

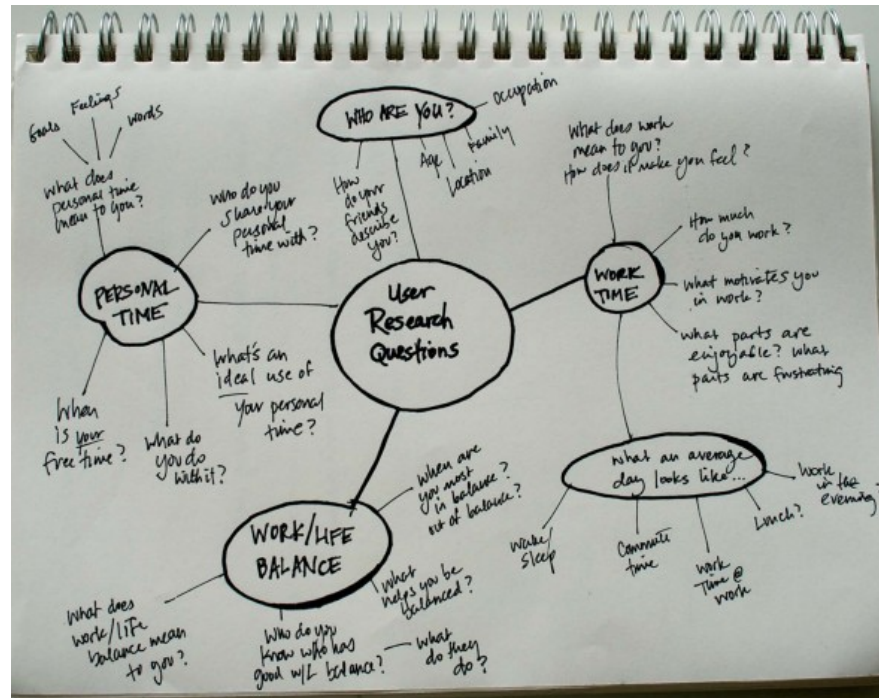
BELIEF	CERTAINTY	NOTES	RESEARCH METHODS	EVIDENCE
Typical User Profile				
Busy Professionals - Our most likely users are busy professionals (e.g., work more than 50 hours a week. Probably skew more female, and many have children/families. Age range is 30s and 40s.	Medium	The high-level details are taken from our typical enterprise user. Need to confirm whether the consumer oriented product would appeal to the same type of user, or someone else.	[+] Doublecheck marketing reports and business case [+] Embed dry test in enterprise product and survey respondents	Marketing Segments
Needs				
Family-Oriented - Users are interested in family-oriented activities.	Low	Purely speculative. We need to do more to understand the variety of activities that users are interested in.	[+] Contextual interviews with 5-8 target users	Personas + Behavioral Dimensions
Stressed by technology - Our typical user feels some stress due to the "always on" nature of technology and devices. It's the desire to eliminate or reduce that stress that leads users on a search for products like ours -- therefore users may be having an experience of stress at the moment they first encounter our product.	Medium	True for enterprise customer; need to confirm with target consumer customer.	[+] Do market research review (especially Pew reports) [+] Contextual interviews with 4-6 target users	Marketing Segments Personas
Motivations				
Achieving and accomplishing - Users are high achievers and interested in accomplishing more outside of work, but not necessarily scaling back what they do at work.	Medium	True for enterprise customer; need to confirm with target consumer customer.	[+] Contextual interviews with 5-8 target users	Personas + Behavioral Dimensions
General goals - Goals tend to be generic: vacation, retirement, get in shape. Not a lot of adventure seekers and bucket listers.	Low	This is an assumption we've always made about enterprise customers, but never really confirmed.	[+] Contextual interviews with 5-8 target users [+] Intercept survey of current users	Personas
Behaviors				
Balance Achieved Through Activities - Ways they achieve a sense of balance now: church or spiritual practices, TV, exercise, commute time (listening to music, books on CD, etc.), shopping, reading print material (books and magazines), FaceBook for viewing photos and catching up with friends.	Medium	True for enterprise customer; need to confirm with target consumer customer.	[+] Contextual interviews with 5-8 target users (include journey lines)	Personas Experience Map Journey Line
Always On - Users are not early technology adopters, but they tend to be online a lot. Usually have computers at home and at work (personal computer at work, shared computer at home). Also likely to have a smart phone, and others in their household are also likely to have smart phones.	Medium	True for enterprise customer; need to confirm with target consumer customer.	[+] Diary study with 2-3 target users	Personas + Behavioral Dimensions

Guerrilla user research

- Guerrilla user research is a fast, low-cost, opportunistic approach to collect insights and feedback from real users by directly interacting with them
- Guerrilla user research is typically based on face to face interaction but can also take other forms (e.g., calling a user on the phone)
- You can use it during envision to clarify who your users really are and what they need and to test if your idea makes sense
 - You can also use it later on to evaluate early design directions

- Start by asking yourself whom you'd like to talk to and how
- Go where your potential users already are:
 - General consumers → cafés, libraries, parks
 - Professionals → coworking spaces, trade fairs, workplaces
 - Students → campuses, public transit, online communities
- You don't need a perfect demographic match - proximity to your user type is often enough for early-stage learning

- Think about what you're trying to learn from your users and list your questions (e.g., do users understand what the app does? would people use this in their daily routine? which of two home screen layouts feels clearer?)



- Go into the field
 - For the research session, ideally meet with the user in his own environment
 - Approach participants politely (e.g., “Hi, I’m working on a new mobile app idea for [topic]. Could I get your quick feedback? It’ll take only a few minutes”)
- Run short, conversational sessions
 - Each session should last no more than 5-10 minutes
 - Explain who you are and what you’re testing (the concept)
 - Describe or show prototype or scenario (without overexplaining)
 - Ask open-ended questions (e.g., “Would you use this app? Why or why not?”)

- Mine the data for insights
 - Spend some time reviewing your notes from all the interviews and looking for the answers to those questions that you listed at the beginning
 - Look for patterns, not individual opinions
 - After 5–10 quick sessions, you'll often have strong directional insight
 - Refine your vision based on what you learn

Competitive assessment (competitor analysis)

- In a competitive assessment, you look at products and services that your target users are likely to encounter and use
 - Direct competitors: apps that solve the same user problem or target the same audience (e.g., Strava, Nike Run Club, or Fitbit for a fitness tracking app)
 - Indirect competitors: apps that solve a related problem or compete for the same user attention/time (e.g., YouTube Fitness channels, Apple Health, or Habit trackers).
- The idea is that people develop their expectations from products and services that they use

- Start by creating a list of the products you want to include in your assessment
 - Focus on products with a larger user base but also check new products
 - Do not include too many products (about 5)
- Decide what you want to learn from your competitors
 - Common goals include understanding how others solve the same user problem; identifying patterns, best practices, and usability norms; finding gaps and differentiation opportunities
- Assess products (horizontally or vertically)

- Collect basic information (name, developer, downloads, rating, ...)
- Identify key features and possibly create a feature comparison table to visualize overlaps and gaps

Feature	App A	App B	App C	Notes
Account creation (email/social)	✓	✓	✗	App C only allows Apple ID
Push notifications	✓	✓	✓	A overuses notifications
Offline mode	✗	✓	✗	Strong advantage for B
Community sharing	✓	✓	✓	All have it, but C integrates better
In-app purchases	✓	✓	✓	Similar pricing tiers

- Analyze the overall quality of the experience offered

Discovery & Onboarding	How easy is it to understand what the app does? Is onboarding simple? Is account creation frictionless?
Core Use / Tasks	How intuitive is task flow? Are navigation patterns consistent? What gestures or interactions stand out?
Visual & Interaction Design	Colors, icons, typography, microinteractions — do they feel modern, clean, cluttered?
Feedback & Guidance	How does the app communicate success, errors, or completion?
Engagement & Retention	Push notifications, progress tracking, rewards, reminders, gamification
Accessibility & Performance	Text size, contrast, speed, offline behavior
Delight Factors	Subtle touches that make the experience memorable (animations, personalization, tone of voice)

- Take screenshots or short recordings to illustrate key UX decisions
- Check user comments on the products (what do users love and hate about them?)
 - Identify recurring themes (e.g., “great tracking,” “buggy,” “confusing login”)
- Visit each competitor’s website, marketing materials, or social media to understand where you can position your app differently

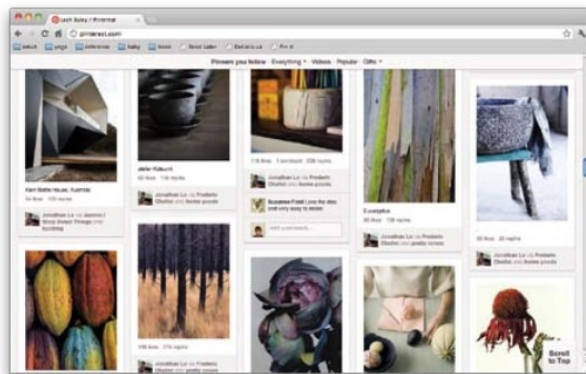
Verdana 10 B I U							Sheets	Charts	SmartArt Graphics	WordArt
1	A	B	C	D	E	F	G			
2	Comparative Assessment									
3	Last Updated July 22									
4										
5	Comparator	Content	Design	Features & Functionality	Continuity	Intuitiveness				
6	Facebook	Entirely comprised of user generated content, with a heavy mix of video, photos, events, etc. Unique timeline layout encourages a chronological reading experience that goes further and further back in time. Commerical content is interspersed seamlessly with content from people you know. Content is typed in a variety of ways (news, wall, photos, status, etc) and that can make it confusing to know where to go to see what in the interface. It does encourage a meandering browsing experience, however, which translates to lots of time on the site.	Busy, cluttered (but that's sort of the point). For the most part, the design is about the content. They do use small icons to denote different types of content, but it doesn't seem to add to the legibility or distinct personality of FB -- if anything, it makes it look a little more unprofessional.	Post public messages Post private messages Upload and share images, videos and other content View other people's posts Create and join groups "Install" apps Create events Respond to events Find music that others are listening to Chat "Friend" someone "Unfriend" someone Update account and privacy settings View ads Create ads Decorate your timeline Search for people, pages, places, groups, apps, events, music, posts, and the internet	Mobile, tablet, and web apps are consistent in terms of content. Features are fairly comparable across devices. Design is a little bit inconsistent between timeline view and wall view. Never quite sure which one you're going to see.	It's a mix. The never-ending linking behavior of content is pretty intuitive and ensures that you can pretty much keep poking around on FB forever if ou want to. On the other hand, the full breadth of available content and, indeed, what you can actually do with FB is not completley intuitive, which could lead to the problem that some users don't use or even know about many of the features that are available to them. Often discovery of new features and apps comes from another user having an automatic updated posted to their wall. This is problematic, however, since it relies to a certain extent on serendipitous discovery, and also because over time all those automatic statuses start to feel like ads that the user becomes blind to.				
	Pinterest	Entirely comprised of user generated content. The main focus is images that are "pinned" to a board. A board appears is a never-ending grid of these high-impact images. Each image is accompanied by a user-written description. Other people's pins and boards create a serendipitous and visually-stimulating browng experience	Beautiful, orderly, heavily influence by gird layout. The content (especially images) are definitely the star of the show here.	"Pin" content (clip from elsewhere and display on one of your boards) Create a board Search Pinterest View other people's pins and boards Invite friends "Like" a pin Additional "goodies" include a "pin it" button for the web browser.	Layout and interactions are extremely consistent, whether in your own board, someone else's, on the mobile app, or on the web site.	The concept of clipping and collecting is immediately intuitive. The home page view aggregates pins from people you follow, and clicking a pin gives you access to a board. The relationship between a pin and a baord is not always clear. Since every view on teh site is essentially an attractive collection of images, it's sometimes difficult to know whether you're looking at someone's board, or an aggregate view such as your				

- Summarize and synthesize insights from the assessment:
 - What interaction patterns or conventions emerge? (e.g., all competitors use a bottom navigation bar; most competitors use gamification)
 - Are there unmet needs? (e.g., no one offers offline sync)
 - What UX elements feel outdated or weak?
 - Which opportunities align with your app's vision?
- Consider documenting key insights from the assessment in a “guidelines” document that you can refer to as you proceed through the design process

Content Best Practices

User generated content has become the norm. Even products and services with strong editorial content include features for users to add content.

Flat text is no longer compelling or sufficient, even if it comes from someone you know. Highly visual displays and novel layouts for content (both editorial and user-generated) appear to be emergent best practices.



BEST PRACTICE

Pinterest uses a unique grid layout that makes photos the star of the show.



BEST PRACTICE

Facebook's timeline layout turns standard user generated content into an addictive stroll down memory lane.



BEST PRACTICE

Real Simple breaks lengthy text up into skimmable chunks or slideshow-style navigation.